

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2025

Course: B.Tech

Branch : Mechanical Engineering

Semester :VIII

Subject Code & Name: BTMEC801A Fundamentals of Automotive Systems

Max Marks: 60

Date:

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.

2. Question No. 1 will be compulsory and include objective-type questions.

3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.

4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.

5. Use of non-programmable scientific calculators is allowed.

6. Assume suitable data wherever necessary and mention it clearly.

				(Level/CO)	Marks	
Q. 1	Objective type questions. (Compulsory Question)				12	
1	The ideal cycle for a petrol engine is.				CO1	1
	a. Diesel cycle	b. Otto cycle	c. Dual cycle	d. Rankine cycle		
2	Which of the following has the highest thermal efficiency?				CO1	1
	a. Two-stroke petrol engine	b. Four-stroke petrol engine	c. Two-stroke diesel engine	d. Four-stroke diesel engine		
3	The clearance volume of an IC engine is				CO1	1
	a. Volume displaced by piston	b. Volume of cylinder	c. Volume above piston at TDC	d. Volume below piston at BDC		
4	The compression ratio of a CI engine is generally				CO2	1
	a. 4:1 to 6:1	b. 6:1 to 8:1	c. 8:1 to 10:1	d. 14:1 to 25:1		
5	The thermal efficiency of a CI engine is higher than SI engine because of				CO2	1
	a. Lower fuel consumption	b. Higher compression ratio	c. Better lubrication	d. Less friction		
6	Exhaust Gas Recirculation (EGR) is mainly used to reduce				CO3	1
	a. CO emissions	b. HC emissions	c. NOx emissions	d. SO ₂ emissions		
7	At which operating condition is Exhaust Gas Recirculation generally				CO3	1

	not used?					
	a. Idle condition	b. Medium load	c. Part load	d. Full load		
8	The purpose of clutch slipping during engagement is to				CO3	1
	a. Increase torque	b. Avoid sudden shock	c. Increase speed	d. Reduce friction		
9	Which type of gearbox is most commonly used in heavy trucks?				CO4	1
	a. Sliding mesh gearbox	b. Constant mesh gearbox	c. Synchromesh gearbox	d. Epicyclic gearbox		
10	The hydraulic pump in a power steering system is driven by				CO4	1
	a. Battery	b. Electric motor	c. Engine crankshaft via belt	d. Steering wheel		
11	Which suspension system is most commonly used in passenger cars?				CO5	1
	a. Rigid axle suspension	b. Leaf spring suspension	c. Air suspension	d. Independent suspension		
12	The MacPherson strut suspension consists of				CO5	1
	a. Leaf spring and axle	b. Coil spring and shock absorber combined	c. Torsion bar and damper	d. Air spring and compressor		
Q. 2	Solve the following.					12
A)	What is an engine block? Explain wet sleeve and dry sleeve inserts with neat sketches				CO1,CO2	6
B)	Explain the loop scavenging process in two-stroke engine with neat schematic diagram				CO1,CO2	6
Q.3	Solve the following.					12
A)	Explain different phases of Combustion in IC engine with neat sketch				CO1, CO2	6
B)	Explain exhaust gas recirculation (EGR) method for controlling the emissions from the engine.				CO1, CO2	6

Q. 4	Solve Any Two of the following.		12
A)	Explain the working of clutch in automotive with neat sketch.	CO3, CO4	6
B)	Explain the construction and working of synchromesh gear box with neat sketch.	CO3, CO4	6
C)	Explain the engine power-torque vs speed characteristics for actual IC engine with suitable performance curves	CO3, CO4	6
Q.5	Solve Any Two of the following.		12
A)	What is antilock braking system (ABS)? Explain working with neat sketch.	CO2, CO3	6
B)	Explain the construction and working of Drum Brake with neat sketch.	CO2, CO3	6
C)	Explain the working of Macpherson Strut Suspension system with neat sketch	CO4, CO5	6
Q. 6	Solve Any Two of the following.		12
A)	Explain the working of hydraulic power steering system with neat sketch	CO4, CO5	6
B)	Describe the various types of front suspension systems	CO4, CO5	6
C)	Differentiate between tyre with tube and tubeless tyre.	CO4, CO5	6
	*** End ***		